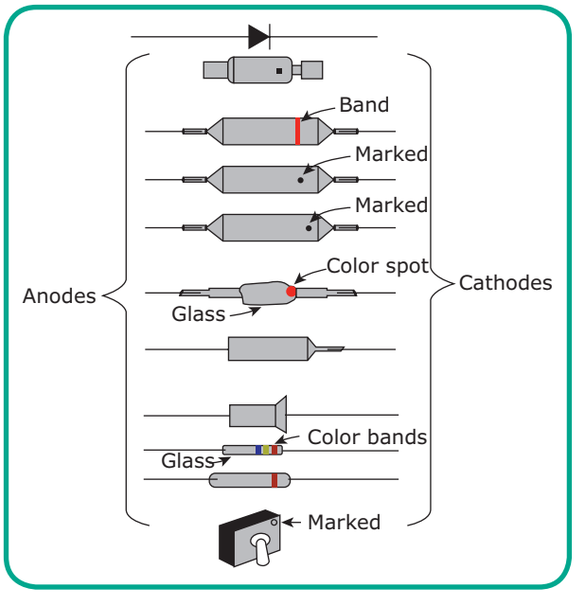


Figure 4.4 **Identification of diodes**

SMD Diode

An SMD diode is a type of diode that is directly soldered on the cooper soldered points in the surface of printed circuit board.

Uses of Diodes

Diodes are used as rectifiers and detectors.

4.3.3 Forward Biasing

To apply forward bias, connect positive terminal of the battery to p-type and negative terminal to n-type as shown in Figure 4.5. Due to this, barrier potential is very much reduced. Positive terminal of the battery repels holes in p-side and negative terminal of the battery repels electrons in n-side. Because of this, current flows in the circuit. This is called "forward current". The magnitude of current depends upon the applied forward voltage.

"When external voltage applied to the junction is in such a direction that it cancels the potential barrier, thus permitting the current flow is called as forward biasing"

The importent term often used with *pn* junction is knee voltage.

4.3.4 Knee Voltage

"It is the forward voltage at which the current through the junction starts to increase rapidly"

The figure 4.6 shows the characteristics of PN junction diode (forward bias).

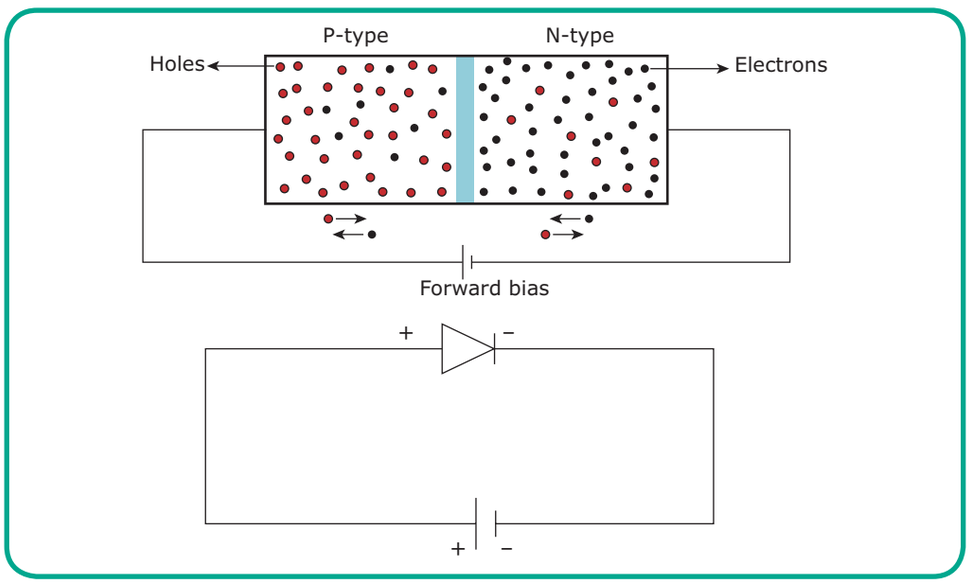


Figure 4.5 **Forward Bias**

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